NVIDIA Earnings Analysis Dashboard

Aniket Gor

March 2, 2026

Executive Summary

Using a Python/Streamlit dashboard, we study NVIDIA's post-earnings announcement behavior via full-period decomposition, event windows (D-20 to D+20), and predictability tests. The full-period results show ~150% total return and ~50% idiosyncratic CAGR, with an outsized share concentrated on the day after earnings release, indicating earnings days have acted as significant repricing events for NVIDIA. Event studies find large D+1 jumps in H1 of 2023 and 2024, and mostly negative post-earnings idiosyncratic returns thereafter. CAR Drift plots and ACF/PACF suggest little persistent PEAD; volatility trends downward since March 2025.

Dashboard link: *nvda-earnings-analyais-app.streamlit.app*

1. APPROACH

This analysis explores the post-earnings-announcement drift (PEAD) for NVIDIA, examining how earnings events impact its stock returns. The accompanying interactive dashboard, built and hosted using Python and Streamlit, allows users to take a macro-to-micro view—identifying high-return periods, analyzing the underlying factor drivers, and exploring pre- and post-earnings event dynamics.

The dashboard is structured to guide the user from a broad historical perspective down to specific event-driven insights. It consists of the following tabs:

- `Full Period Analysis` — A macro view displaying NVIDIA's returns, risk metrics, and factor decompositions over the entire period.
- `Earnings Events` — An event-focused view analyzing cumulative abnormal returns around earnings dates (D-20 to D+20).
- `Predictability` — An investigative view testing the relationship between pre- and post-earnings returns.
- `Data Explorer` — A transparency view providing summary statistics and sanity checks on the underlying data.

The dashboard is designed to facilitate iterative exploration:

Macro View (Full Period Analysis): Users begin with total returns broken down into systematic and idiosyncratic components. They can view annual returns, drill down into specific years (e.g., 2023), and explore daily time series, drawdowns, compounded dollar values, and factor contributions. Earnings events are overlaid on these charts, enabling users to visually pinpoint where NVIDIA's returns were concentrated.

Event View (Earnings Events): Focusing on a 41-day window surrounding the earnings announcement (D-20 to D+20), users can analyze whether returns were driven by pre-event anticipation or post-event drift. Interactive heatmaps highlight significant return days, while time-series charts illustrate the presence or absence of a classic PEAD. Factor decomposition is also provided for these specific event windows.

Predictability and Data Sanity: Users can then investigate potential correlations between pre- and post-event returns. Finally, the Data Explorer provides raw data validation, displaying distributions and key statistics (mean, median, max, standard deviation) to ensure data integrity.

1.1. Convention and Definitions

- Day 0: The date of the earnings announcement, assuming the release occurs after the market close.
- Day+1: The first full trading day following the earnings announcement.
- Idiosyncratic Return: Calculated for a given day t as NVIDIA's total daily return minus the systematic return (derived from the dot product of the stock's factor exposures and the corresponding factor returns).
- Cumulative Abnormal Returns (CAR): In alignment with standard academic literature, we calculate CAR using an additive approach rather than multiplicative compounding. While the difference between the two methods is generally negligible over short windows with low returns, high volatility can magnify the gap. Although multiplicative compounding more accurately reflects actual portfolio growth, additive aggregation is utilized here because it simplifies multi-period performance attribution.
- Pre-Event Window (D-20 to D0): Analyzes the price "run-up" or anticipatory information leakage during the 20 trading days leading up to the earnings announcement.
- Post-Event Window (D+1 to D+20): Defines the Post-Earnings Announcement Drift (PEAD) observation window. This period measures the magnitude and persistence of the drift over the 20 trading days following the release.

2. LITERATURE

In a perfectly efficient market, asset prices should ideally respond to earnings surprises. However, historical evidence demonstrates a systemic under-reaction to these announcements, leading to a momentum effect known as Post-Earnings Announcement Drift (PEAD). Bernard and Thomas (1989) established foundational evidence for this phenomenon, showing that markets are slow to fully absorb earnings shocks.

The persistence of PEAD is primarily attributed to sticky expectations—where analysts are slow to iteratively update their beliefs—and broader investor inattention. The academic literature highlights how the context of information dissemination significantly impacts market reaction. Fedyk (2018) documents a "headline bias," where front-page news commands a stronger market reaction compared to information buried deeper in reports or terminals. Similarly, DellaVigna and Pollet (2009) found stronger inattention effects during Friday announcements, while Peress (2008) showed that prominent coverage (such as in the WSJ) correlates with high investor attention.

Recent market data indicates that classic PEAD is broadly disappearing due to increased market efficiency. Historically, PEAD has been much stronger in small-cap equities than in large caps (Bernard and Thomas, 1989). Consequently, for a heavily scrutinized mega-cap company like NVIDIA, traditional PEAD should theoretically be weak or non-existent. However, the rise of the AI industry represents uncharted territory where analyst forecast errors regarding revenue and earnings growth could be large and persistent enough to drive meaningful post-earnings drift, making NVIDIA a unique case study for testing the modern presence of PEAD.

3. INSIGHTS AND FINDINGS

3.1. Full Period Analysis

Over the analyzed period, NVIDIA exhibited a total Compound Annual Growth Rate (CAGR) of approximately 150%. A factor decomposition of this performance reveals that roughly a third of the total return was idiosyncratic, while approximately two-third was driven by systematic factors. Most notably, the trading day immediately following earnings announcements (D+1) accounted for an outsized portion of the idiosyncratic gains. Cumulatively, D+1 returns accounted for 40% of the 49% idiosyncratic return.

Year-over-Year Concentration NVIDIA's returns were highly concentrated in 2023 and 2024. The first two quarters of both 2023 and 2024 contributed heavily to the stock's overall growth:

Q1 and Q2 for both years saw massive single-day jumps of 10% to 20% on earnings release days. Total quarter gains were split between idiosyncratic returns and systematic drivers—specifically the Semiconductors, Market, and Beta factors. **Interestingly, idiosyncratic returns have disappeared since 2025.**

Drawdowns and Non-Earnings Volatility

Majority of NVIDIA's drawdowns occurred between earnings events rather than on the announcement dates themselves. For example, the maximum drawdown period took place between January and July 2025, largely independent of earnings releases.

3.2. Earnings Events

During the early pre-earnings window (D-20 to D-10), NVIDIA generally exhibits positive idiosyncratic returns. However, as the announcement date approaches (D-5 to D0), idiosyncratic returns shrink in magnitude and display a mix of positive and negative daily movements.

Large idiosyncratic returns are heavily concentrated precisely on the D+1 day, specifically during the first two quarters of both 2023 and 2024 (highlighted by single-day spikes of 8% to 16%). Post Q2 of 2024, D+1 idiosyncratic returns have skewed negative.

The time-series analysis indicates a general absence of a persistent, classic PEAD. On average, NVIDIA's stock experiences a discrete step-function jump—either up or down—on D+1 and remains relatively flat thereafter. However, a quarter-by-quarter breakdown reveals nuanced behaviors:

- 2023 Q1 & Q2: The stock experienced massive jumps on D+1 and immediately stabilized, exhibiting zero post-earnings drift.
- 2024 Q1 & Q2: A slight positive drift was observable following the D+1 jump, persisting for up to 10 days post-earnings (D+10) before flattening out.
- 2024 Q3 Onwards: The trend inverted. While pre-earnings returns remained generally positive (indicative of anticipatory run-ups), the majority of post-earnings returns from 2024 Q3 onward have been decidedly negative.
- 2025 Q3: We see PEAD having a persistent negative drift.

Factor Composition Around Earnings

Examining the factor decomposition during these event windows provides further insights on the underlying drivers of these returns:

- 2023 Q1 & Q2: The massive post-earnings gains were primarily driven by the idiosyncratic and semiconductors factor returns, supplemented by standard Market and Beta factor exposures.
- 2024 Q1 & Q2: Alongside the core factors, Momentum and Growth factors emerged as noticeable contributors.
- 2024 Q3 (Downside Regime): During this negative post-earnings period, Total Returns, Idiosyncratic Returns, Semiconductors, and Momentum factors all turned distinctly negative.
- 2025 Q1 & Q3: Interestingly, while NVIDIA's idiosyncratic returns remained negative during these periods, broad systematic factors—specifically Market, Beta, and Momentum—were positive.

3.3. Predictability

An analysis of the metric correlation matrix shows that total returns exhibit a distinct contrarian dynamic around the event: pre-earnings Cumulative Abnormal Returns (CAR) are negatively correlated with both post-earnings CAR and post-earnings volatility.

Volatility Time Series Trends

The big single-day return movements during the first two quarters of 2023 and 2024 generated significant volatility spikes. However, idiosyncratic and total returns volatility has been exhibiting a clear and consistent downtrend since March 2025 up until September 2025, suggesting that the market is increasingly pricing in NVIDIA's earnings outcomes with less post-release turbulence.

Autocorrelation and Persistence Check

The time-series models show that idiosyncratic returns behave like random walk and do not indicate any structural mean reversion or persistent momentum trends, confirming the visual absence of a classic, prolonged Post-Earnings Announcement Drift (PEAD).

4. CONCLUSION

In conclusion, this analysis demonstrates that despite the unprecedented growth and uncharted territory of the AI sector, NVIDIA does not exhibit a classic, persistent Post-Earnings Announcement Drift (PEAD). Instead, the market processes these earnings shocks with remarkable efficiency. The vast majority of price discovery occurs as a discrete step-function jump on the immediate post-earnings day (D+1), which historically accounts for nearly half of the stock's total idiosyncratic returns.

Furthermore, the data reveals a regime shift over the analyzed period. While 2023 and early 2024 were defined by massive positive idiosyncratic shocks and high event-driven volatility, the landscape from late 2024 through 2025 shows diminishing idiosyncratic returns and a consistent downtrend in post-release turbulence. Supported by autocorrelation models confirming that post-event returns behave as a random walk, the evidence suggests that the market has become increasingly adept at swiftly pricing in NVIDIA's earnings outcomes, effectively neutralizing any tradable momentum or prolonged drift.

DASHBOARD LINK

nvda-earnings-analyais-app.streamlit.app

GITHUB REPOSITORY

Visit the repository to access the source code.

https://github.com/aniket-gor/nvda-earnings-analyais-app

CONTACT ME

In case of any feedback or queries, reach out:

aniket.gor@berkeley.edu
www.linkedin.com/in/aniket-gor/